Physics 1st Paper - Chapter 02 - Vector Problem Book

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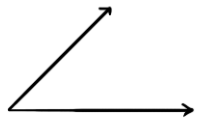
1 Finding the Magnitude and Direction of the Resultant of two Vectors

1.1 Geometric Addition and Subtraction

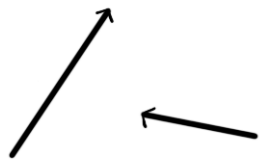
1.1.1

Find (a) $\vec{a}+\vec{b}$, (b) $\vec{a}-\vec{b}$ and (c) $\vec{b}-\vec{a}$



1.1.2

Find (a) $\vec{a}+\vec{b}$, (b) $\vec{a}-\vec{b}$ and (c) $\vec{b}-\vec{a}$



1.1.3

The magnitudes of the vectors $\vec{A}$, $\vec{B}$, $\vec{C}$ are respectively 12, 5 and 13, and $\vec{A}+\vec{B}=\vec{C}$. What is the angle between $\vec{A}$ and $\vec{B}$.

1.1.4

Consider two displacements, one of magnitude $3m$ and another of magnitude $4m$. Show how the displacement vectors may be combined to get a resultant displacement of magnitude (a) $7m$, (b) $1m$, and (c) $5m$.

1.2 Numerical Questions on Addition

1.2.1

The angle between two vector quantities with magnitude $20N$ and $60N$ is $30°$. What is the magnitude and direction of their resultant.

1.2.2

Two vectors having equal magnitude A make an angle θ with each other. Find the magnitude and direction of the resultant.

1.2.3

Two vectors have magnitude 5 units. They act at an angle $120°$ through the same point. Find the magnitude and direction of their resultant.

1.2.4

Two vectors of equal magnitude 5 unit have an angle 60° between them. Find the magnitude of the sum of the vectors.

1.2.5

Two vectors of equal magnitude are acting at a point. The magnitude of the resultant is equal to the magnitude of any one of them. Determine the angle between them.

1.2.6

Maximum value of the resultant of two vectors is 28 and minimum value is 4 units. How do they act if the resultant is 5 times the minimum value of the resultant?

1.2.7

A body is pulled towards east with $50N$ force and then it is pulled towards north with $20N$ force making an angle 60° with the east. Find the magnitude and direction of their resultant force.

1.2.8

Quvi travels a distance of $250km$ at angle 30° towards South-East by motorcycle. Then he travels $250km$ at angle 30° towards South-West. What is the value and direction of the displacement vector of Quvi?

1.2.9

A vector $\vec{A}$ makes an angle of 20° and $\vec{B}$ makes and angle of 110° with the X-axis. The magnitudes of these vectors are $3m$ and $4m$ respectively. Find the resultant.

1.2.10

Let $\vec{A}$ and $\vec{B}$ be the two vectors of magnitude 10 unit each. If they are inclined to the X-axis at angles 30° and 60° respectively, find the resultant.

1.2.11

Two vectors have magnitudes 3 unit and 4 unit respectively. What should be the angle between them if the magnitude of the resultant is (a) 1 unit, (b) 5 unit and (c) 7 unit.

1.3 Numerical Questions on Subtraction

1.3.1

Two vectors of equal magnitude 5 unit have an angle 60° between them. Find the magnitude of difference of the vectors.

1.4 Resultant in Rectangular Coordinate System

1.4.1

Let $\vec{a} = 4\hat{i} + 3\hat{j}$ and $\vec{b} = 3\hat{i} + 4\hat{j}$. Find the magnitudes of (a) $\vec{a}$, (b) $\vec{b}$, (c) $\vec{a} + \vec{b}$ and (d) $\vec{a} - \vec{b}$.

1.4.2

The position vectors of P and Q are $P(\hat{i} + \hat{j} + \hat{k})$ and $Q(2\hat{i} + 3\hat{j} - 2\hat{k})$. Calculate the vector $\vec{PQ}$. What is the magnitude of this vector?

1.4.3

A ship sets out to sail to a point $120km$ due north. An unexpected storm blows the ship to a point $100km$ due east of its starting point. (a) How far and (b) in what direction must it now sail to reach its original destination?

1.4.4

If $\vec{B}$ is added to $\vec{C} = 3\hat{i} + 4\hat{j}$, the resultant acts along the positive direction of y-axis and the magnitude of the resultant is equal to the magnitude of $\vec{C}$. $\vec{B} =?$

2 Finding Orthogonal Projections and Perpendicular Components of a Vector

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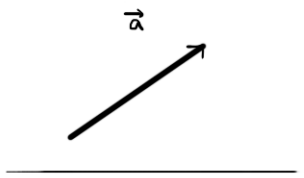
2.1 Applying Geometry

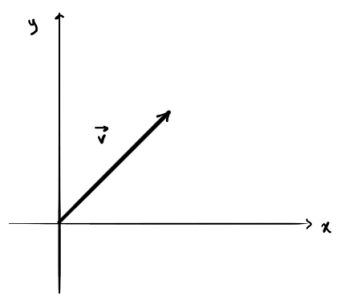
2.1.1

What is the projection and component of the vector along the line in the figure?



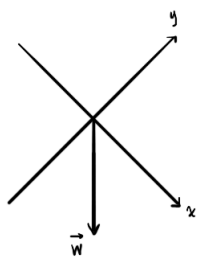
2.1.2

Find the projection and component of the vector in the figure along x and y axis.



2.1.3

Find the projection and component of the vector in the figure along x and y axis.



2.2 Numerical Questions

2.2.1

A force of $200N$ is applied at an angle of $60°$ to the horizontal. What is the horizontal component of the force?

2.2.2

What are (a) the x component and (b) the y component of a vector $\vec{a}$ in the xy plane if its direction is $250°$ counterclockwise from the positive direction of the x axis and its magnitude is $7.3m$?

2.2.3

A displacement vector $\vec{r}$ in the xy plane is $15m$ long and directed at angle $\theta = 30°$. Determine (a) the x component and (b) the y component of the vector.

2.2.4

A force of $10.5N$ acts on a particle along a direction making an angle $37°$ with the vertical. Find the component of the force in the vertical direction.

2.2.5

Wind is blowing with a velocity of $20km/h$ towards the south making an angle $60°$ with the west. What are the magnitudes of the component of the wind velocity towards north and east direction?

2.2.6

A body is pulled towards the west by the force of $10N$ making an angle $30°$ with the south. What are the magnitudes of the south and west direction components of the force?

2.2.7

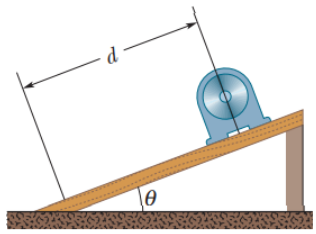
A small airplane leaves an airport on an overcast day and is later sighted $215km$ away, in a direction making an angle of $22°$ east of due north. This means that the direction is not due north (directly toward the north) but is rotated $22°$ toward the east from due north. How far east and north is the airplane from the airport when sighted?

2.2.8

The x component of vector $\vec{A}$ is $-25.0m$ and the y component is $+40.0m$. (a) What is the magnitude of $\vec{A}$? (b) What is the angle between the direction of $\vec{A}$ and the positive direction of x?

2.2.9

In the fig, a heavy piece of machinery is raised by sliding it a distance $d = 12.5m$ along a plank oriented at angle $\theta = 20.0°$ to the horizontal. How far is it moved (a) vertically and (b) horizontally?



2.3 Orthogonal Projections and Perpendicular Components in Rectangular Coordinate System

2.3.1

A vector has component along the X-axis equal to 25 unit and along the Y-axis equal to 60 unit. Find the magnitude and direction of the vector.

2.4 Lawn Roller

2.4.1

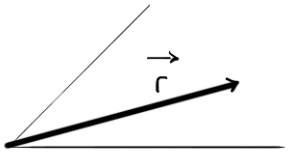
While pushing or pulling a lawn roller, you are applying a force of $19.6N$ making an angle $30°$ with the horizontal. It is easier to pull than to push because while pulling its weight reduces by ?

3 Finding Components of a Vector (Generic)

3.1 Applying Geometry

3.1.1

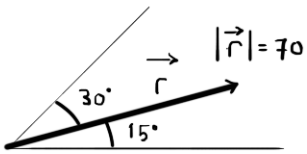
Find the two components of the vector in the figure along the two lines.



3.2 Numerical Questions

3.2.1

Find the two components of the vector in the figure along the two lines.



4 Magnitude and Unit Vector of a Vector in Rectangular Coordinate System

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4.0.1

Write the magnitude and the unit vector in the direction of $\vec{A} = 5\hat{i} + \hat{j} - 2\hat{k}$.

4.0.2

If $\vec{R} = 4\hat{i} - 6\hat{j} + 12\hat{k}$, then find its parallel unit vector.

4.0.3

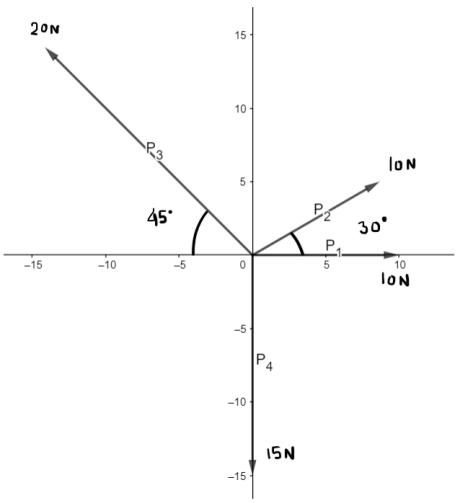
If $\vec{r_1} = 2\hat{i} + 4\hat{j} - 5\hat{k}$ and $\vec{r_2} = \hat{i} + 2\hat{j} + 3\hat{k}$, find the parallel unit vector of the resultant of these two vectors.

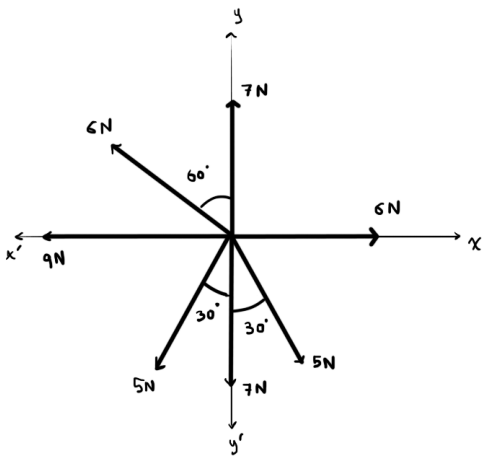
5 Finding the Magnitude and Direction of the Resultant of Multiple Vectors

5.1 Numerical Questions

5.1.1

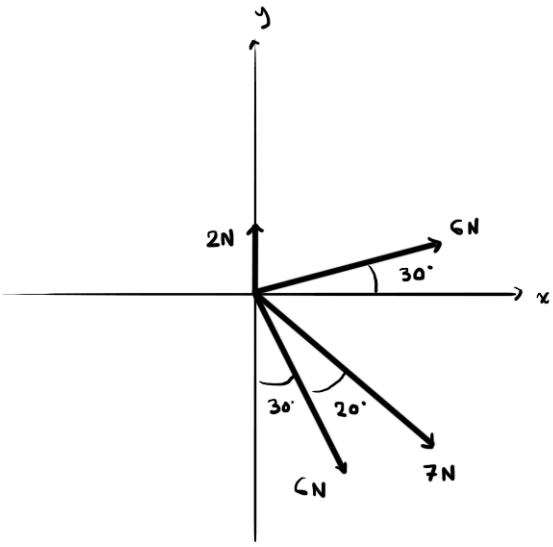
Find the resultant of the following vectors:

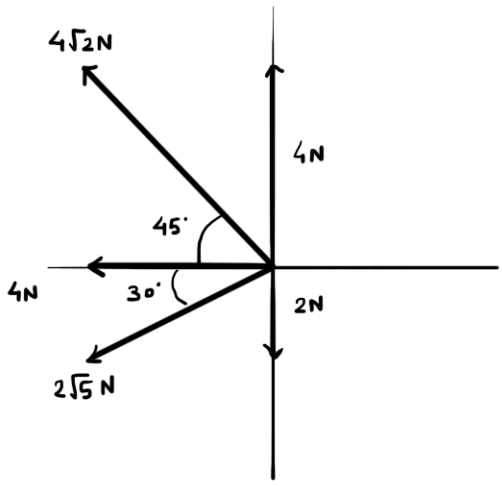




5.1.2

Find the resultant of the following vectors:





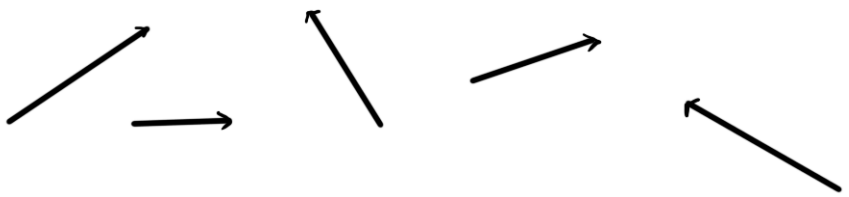
5.1.3

Add vectors $\vec{A}, \vec{B}$ and $\vec{C}$ each having magnitude of 100 unit and inclined to the X-axis at angles 45°, 135° and 315° respectively.

5.2 Applying Geometry

5.2.1

Find the resultant of the following vectors:



5.3 Resultant in Rectangular Coordinate System

5.3.1

The following four forces are acting on a particle. Calculate the resultant force. What will be the magnitude of the resultant force?
$\vec{F_1} = 2\hat{i} + 3\hat{j} - 5\hat{k}, \vec{F_2} = -5\hat{i} + \hat{j} + 3\hat{k}, \vec{F_3} = \hat{i} + 2\hat{j} + 4\hat{k}, \vec{F_4} = 4\hat{i} - 3\hat{j} - 2\hat{k}.$

5.3.2

$\vec{A} = 3\hat{i} - 2\hat{j} + \hat{k}, \vec{B} = 2\hat{i} - 4\hat{j} - 3\hat{k}, \vec{C} = -\hat{i} + 2\hat{j} + 2\hat{k}$
(i) $|\vec{C}| =?$, (ii) $|\vec{A} + \vec{B} + \vec{C}| =?$, (iii) $|2\vec{A} - 2\vec{B} - 5\vec{C}| =?$

5.3.3

If $\vec{B}$ is added to $\vec{C} = 3.0\hat{i} + 4.0\hat{j}$, the result is a vector in the positive direction of the y axis, with a magnitude equal to that of $\vec{C}$. What is the magnitude of $\vec{B}$?

6 Scalar Product of two Vectors / Vector Dot Product

6.1 Using Magnitudes and Angle

6.1.1

Vectors $\vec{C}$ and $\vec{D}$ have magnitudes of 3 units and 4 units, respectively. What is the angle between the directions of $\vec{C}$ and $\vec{D}$ if $\vec{C}.\vec{D}$ equals (a) zero, (b) 12 units, and (c) -12 units?

6.1.2

The work done by a force $\vec{F}$ during a displacement $\vec{r}$ is given by $\vec{F}.\vec{r}$. Suppose a force of $12N$ acts on a particle in vertically upward direction and the particle is displaced through $2.0m$ in vertically downward direction. Find the work done by the force during this displacement.

6.1.3

What is the value of $\vec{A}.\vec{A}$?

6.2 Dot Product in Rectangular Coordinate System

6.2.1

If $\vec{A} = 6\hat{i} - 3\hat{j} + 2\hat{k}$ and $\vec{B} = 2\hat{i} + 2\hat{j} + \hat{k}$, then find $\vec{A}.\vec{B}$.

6.2.2

If $\vec{A} = 2\hat{i} + 4\hat{j} + 2\hat{k}$ and $\vec{B} = \hat{i} - 2\hat{j} + 4\hat{k}$, then find $(\vec{A} - \vec{B}).(\vec{A} + \vec{B})$.

6.2.3

A force acting $\vec{F} = (6\hat{i} - 3\hat{j} + 2\hat{k})N$ acting on a particle displaces it by $\vec{r} = (2\hat{i} + 2\hat{j} - 2\hat{k})m$, find the work done by the force.

7 Perpendicularity/Orthogonality of two Vectors

7.0.1

$\vec{A} = 2\hat{i} + 4\hat{i} + 7\hat{k}$ and $\vec{B} = 3\hat{i} - 5\hat{j} + 2\hat{k}$. Show that, these two vectors are at right angle to each other.

7.0.2

Find a, if $\vec{A} = 3\hat{i} + \hat{j} - a\hat{k}$ and $\vec{B} = 4\hat{i} - a\hat{j} + a\hat{k}$ are perpendicular to each other.

7.0.3

$\vec{A} = 2\hat{i} + 3\hat{j} - 5\hat{k}$ and $\vec{B} = m\hat{i} + 2\hat{j} + 10\hat{k}$, for what value of m, will these two vectors will be perpendicular to each other?

7.0.4

For what value of a, are $\vec{A} = 2a\hat{i} + a\hat{j} - 4\hat{k}$ and $\vec{B} = a\hat{i} - 2\hat{j} + \hat{k}$ perpendicular to each other?

8 Finding Orthogonal Projection and Component of a Vector along another Vector

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8.0.1

Find the orthogonal projection of $\vec{A}$ on $\vec{B}$ if $\vec{A} = 2\hat{i} + 2\hat{j} - \hat{k}$ and $vecB = 2\hat{i} + \hat{j} - 2\hat{k}$.

8.0.2

If $\vec{A} = 3\hat{i} - \hat{j} + 3\hat{k}$ and $\vec{B} = \hat{i} - 4\hat{j} + 2\hat{k}$, then calculate the orthogonal projection of $\vec{B}$ on $\vec{A}$ and the component of $\vec{B}$ along $\vec{A}$.

8.0.3

If $\vec{A} = 9\hat{i} + \hat{j} - 6\hat{k}$ and $\vec{B} = 4\hat{i} - 6\hat{j} + 5\hat{k}$, find the projection of $\vec{A}$ on $\vec{B}$ and projection of $\vec{B}$ on $\vec{A}$.

9 Finding the Angle between two Vectors

9.0.1

Vector $\vec{A}$ has a magnitude of 6.00 units, vector $\vec{B}$ has a magnitude of 7.00 units, and $\vec{A}.\vec{B}$ has a value of 14.0. What is the angle between the directions of $\vec{A}$ and $\vec{B}$?

9.0.2

What is the angle ϕ between $\vec{a} = 3.0\hat{i} - 4.0\hat{j}$ and $\vec{b} = -2.0\hat{i} + 3.0\hat{k}$?

9.0.3

Let $\vec{a} = 2\hat{i} + 3\hat{j} + 4\hat{k}$ and $\vec{b} = 3\hat{i} + 4\hat{j} + 5\hat{k}$. Find the angle between them.

9.0.4

If $\vec{a} = 2\hat{i} + 3\hat{j} + 4\hat{k}$ and $\vec{b} = 4\hat{i} + 3\hat{j} + 2\hat{k}$, find the angle between $\vec{a}$ and $\vec{b}$.

9.0.5

Find the angle between $\vec{A}$ and $\vec{B}$ if $\vec{A} = 4\hat{i} + 3\hat{j} - 5\hat{k}$ and $\vec{B} = 2\hat{i} + \hat{j} + 3\hat{k}$.

9.0.6

Find the angle between the vectors if $|\vec{A} + \vec{B}| = |\vec{A} - \vec{B}|$.

9.0.7

Two cars A and B. The vertical projections of their velocities are $\sqrt{3}m/s$ and $2m/s$ respectively and the horizontal projections are $1m/s$ and $2m/s$, What is the angle between them?

10 Finding the Direction Cosines of a Vector

10.0.1

If $\vec{A} = 5\hat{i} - 6\hat{j} + 7\hat{k}$, find the direction cosines of the vector.

11 Finding the Angles of a Vector with the Axes

11.1 2D Coordinate System

11.1.1

(a) What angle does the vector $\vec{u} = 3\hat{i} + 4\hat{j}$ make with x and y axis?
(b) What angle does the vector $\vec{u} = -3\hat{i} + 4\hat{j}$ make with x and y axis?
(c) What angle does the vector $\vec{u} = -3\hat{i} - 4\hat{j}$ make with x and y axis?
(d) What angle does the vector $\vec{u} = 3\hat{i} - 4\hat{j}$ make with x and y axis?

11.2 3D Coordinate System

11.2.1

If $\vec{A} = 5\hat{i} - 6\hat{j} + 7\hat{k}$, find the angles of the vector with the axes.

11.2.2

If $\vec{A} = 2\hat{i} + 2\hat{j} - \hat{k}$, what is the angular displacement of $\vec{A}$ from X-axis?

11.2.3

Can you find the angle that $A = 3\hat{i} + 4\hat{j} + 6\hat{k}$ forms with the $X - Y$ plane? If yes, inform me the answer and collect a gift.

12 Vector Product of two Vectors / Vector Cross Product

12.1 Using Magnitudes and Angle

12.1.1

Vectors $\vec{C}$ and $\vec{D}$ have magnitudes of 3 units and 4 units, respectively. What is the angle between the directions of $\vec{C}$ and $\vec{D}$ if the magnitude of the vector product $\vec{C} \times \vec{D}$ is (a) zero, (b) 12 units?

12.1.2

The vector $\vec{A}$ has a magnitude of 5 unit, $\vec{B}$ has a magnitude of 6 unit and the cross product of $\vec{A}$ and $\vec{B}$ has a magnitude of 15 unit. Find the angle between $\vec{A}$ and $\vec{B}$.

12.1.3

What is the value of $\vec{A} \times \vec{A}$?

12.1.4

Prove that, $(\vec{A}.\vec{B})^2 + |\vec{A} \times \vec{B}|^2 = A^2B^2$

12.1.5

Vector $\vec{a}$ lies in the xy plane, has a magnitude of 18 units, and points in a direction 250° from the positive direction of the x axis. Also, vector $\vec{b}$ has a magnitude of 12 units and points in the positive direction of the z axis. What is the vector product $\vec{c} = \vec{a} \times \vec{b}$?

12.1.6

What is the magnitude of $\vec{a} \times (\vec{b} \times \vec{a})$ if $a = 3.90, b = 2.70$, and the angle between the two vectors is 63.0°?

12.2 Cross Product in Rectangular Coordinate System

12.2.1

If $\vec{a} = 3\hat{i} - 4\hat{j}$ and $\vec{b} = -2\hat{i} + 3\hat{k}$, what is $\vec{c} = \vec{a} \times \vec{b}$?

12.2.2

If $\vec{A} = 2\hat{i} + 3\hat{j} + 4\hat{k}$ and $\vec{B} = 4\hat{i} + 3\hat{j} + 2\hat{k}$, find $\vec{A} \times \vec{B}$.

12.2.3

If $\vec{A} = 2\hat{i} - 3\hat{j} + 7\hat{k}, \vec{B} = \hat{i} + 2\hat{k}$ and $\vec{C} = \hat{j} - \hat{k}$, find $\vec{A}.(\vec{B} \times \vec{C})$.

12.2.4

$\vec{A} = \hat{i} - \hat{j} + \hat{k}$ and $\vec{B} = 2\hat{i} - 3\hat{j} + 6\hat{k}$. Calculate the vector product and the angle between these two vectors.

12.2.5

Find the result of the following: $(2\hat{i} - 3\hat{j}).((\hat{i} + \hat{j} - \hat{k}) \times (3\hat{i} - \hat{k}))$

12.2.6

For the following three vectors, what is $3\vec{C}.(2\vec{A} \times \vec{B})$?
$\vec{A} = 2.00\hat{i} + 3.00\hat{j} - 4.00\hat{k}$, $\vec{B} = -3.00\hat{i} + 4.00\hat{j} + 2.00\hat{k}$, $\vec{C} = 7.00\hat{i} - 8.00\hat{j}$

12.2.7

If $\vec{A} = 3\hat{i} + 2\hat{j} + \hat{k}, \vec{B} = \hat{i} = 2\hat{j} - 3\hat{k}$ and $\vec{C} = \hat{i} + \hat{j} + 2\hat{k}$, then prove that, $\vec{A}.(\vec{B} \times \vec{C}) = (\vec{A} \times \vec{B}).\vec{C}$

12.2.8

If $\vec{A} = 2\hat{i} + 4\hat{j} + 2\hat{k}$ and $\vec{B} = \hat{i} - 2\hat{j} + 4\hat{k}$, then find $(\vec{B} - \vec{A}) \times (\vec{B} + \vec{A})$.

12.2.9

If $\vec{A} = \hat{i} + 3\hat{j} + 2\hat{k}, \vec{B} = 3\hat{i} + 2\hat{j} - \hat{k}$ and $\vec{C} = 2\hat{i} - 3\hat{j} + 4\hat{k}$, then prove that, $(\vec{B} + \vec{C}) \times \vec{A} = \vec{B} \times \vec{A} + \vec{C} \times \vec{A}$.

12.2.10

If $\vec{A} = 2\hat{i} - 13\hat{j} + 74\hat{k}, \vec{B} = 9\hat{i} + 29\hat{j} - 45\hat{k}, \vec{A}.(\vec{B} \times \vec{A}) =?$

12.2.11

In the product $\vec{F} = q\vec{v} \times \vec{B}$, take $q = 2$, $\vec{v} = 2.0\hat{i} + 4.0\hat{j} + 6.0\hat{k}$ and $\vec{F} = 4.0\hat{i} - 20\hat{j} + 12\hat{k}$. What then is $\vec{B}$ in unit-vector notation if $B_x = B_y$?

12.2.12

Two vectors are given by $\vec{a} = 3.0\hat{i} + 5.0\hat{j}$ and $\vec{b} = 2.0\hat{i} + 4.0\hat{j}$. Find (a) $\vec{a} \times \vec{b}$, (b) $\vec{a}.\vec{b}$, (c) $(\vec{a} + \vec{b}).\vec{b}$, and (d) the component of $\vec{a}$ along the direction of $\vec{b}$.

12.2.13

Three vectors are given by $\vec{a} = 3.0\hat{i} + 3.0\hat{j} - 2.0\hat{k}, \vec{b} = -1.0\hat{i} - 4.0\hat{j} + 2.0\hat{k}$, and $\vec{c} = 2.0\hat{i} + 2.0\hat{j} + 1.0\hat{k}$. Find (a) $\vec{a}.(\vec{b} \times \vec{c})$, (b) $\vec{a}.(\vec{b} + \vec{c})$, and (c) $\vec{a} \times (\vec{b} + \vec{c})$.

12.2.14

If $\vec{d_1} = 3\hat{i} - 2\hat{j} + 4\hat{k}$ and $\vec{d_2} = -5\hat{i} + 2\hat{j} - \hat{k}$, then what is $(\vec{d_1} + \vec{d_2}).(\vec{d_1} \times 4\vec{d_2})$?

13 Finding the Unit Normal Vectors of a Plane formed by two Vectors

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13.0.1

If $\vec{A} = 2\hat{i} + 4\hat{j} - 5\hat{k}$ and $\vec{B} = \hat{i} + 2\hat{j} + 3\hat{k}$, find the unit vector perpendicular to the plane of $\vec{A}$ and $\vec{B}$.

13.0.2

$\vec{A} = 2\hat{i} + 2\hat{j} - \hat{k}$ and $\vec{B} = 6\hat{i} - 3\hat{j} + 2\hat{k}$ are two vectors. Find the unit vector along their perpendicular direction.

13.0.3

Find the perpendicular unit vector on the plane which is constructed by the two vectors $\vec{A} = 2\hat{i} - 6\hat{j} - 3\hat{k}$ and $\vec{B} = 4\hat{i} + 3\hat{j} - \hat{k}$.

13.0.4

$\vec{A} = \hat{i} - \hat{k}$ and $\vec{B} = \hat{i} + \hat{j}$. Calculate the unit normal vector to the plane formed by these two vectors.

13.0.5

$\vec{P} = 2\hat{i} + 3\hat{j} - 4\hat{k}$ and $\vec{Q} = -2\hat{i} + \hat{j} + 3\hat{k}$. Calculate the unit vector towards the perpendicular direction of the plane on which these two vectors are placed.

14 Parallelism of two Vectors

14.0.1

Show that, $\vec{A} = \hat{i} + \hat{j} + \hat{k}$ and $\vec{B} = 5\hat{i} + 5\hat{j} + 5\hat{k}$ two vectors are parallel to each other.

14.0.2

Show that, $\vec{P} = \hat{i} - 2\hat{j} + \hat{k}$ and $\vec{Q} = 3\hat{i} - 6\hat{j} + 3\hat{k}$ two vectors are parallel to each other.

14.0.3

For what value of α will $\vec{A} = 5\hat{i} + 2\hat{j} - 3\hat{k}$ and $\vec{B} = 15\hat{i} + \alpha\hat{j} - 9\hat{k}$ be parallel to each other?

14.0.4

$\vec{A} = 2\hat{i} + m\hat{j} + n\hat{k}, \vec{B} = 4\hat{i} + 6\hat{j} + 8\hat{k}$ - for what values of m and n will the vectors $\vec{A}$ and $\vec{B}$ be parallel to each other?

14.0.5

If $\vec{P} = 2\hat{i} + m\hat{j} - 3\hat{k}$ and $\vec{Q} = 10\hat{i} - 5\hat{j} - 15\hat{k}$ are parallel to each other, find the value of m.

15 Finding the Equation of a Plane formed by two Vectors

15.0.1

$\vec{A} = 2\hat{i} + 2\hat{j} - \hat{k}$ and $\vec{B} = 6\hat{i} - 3\hat{j} + 2\hat{k}$ are two vectors. Find the equation of the plane formed by these two vectors.

16 Finding Area

16.1 Triangle

16.1.1

$\vec{A} = 4\hat{i} + 3\hat{j} + \hat{k}$ and $\vec{B} = 2\hat{i} + \hat{j} + 5\hat{k}$. If $\vec{A}$ and $\vec{B}$ represent two adjacent sides of a triangle, then what is the area of the triangle?

16.1.2

Find the area of the triangle if the vertices are $(2, 3, 1), (1, 1, 3)$ and $(2, 2, 5)$.

16.2 Parallelogram (Adjacent Sides)

16.2.1

Find the area of a parallelogram whose adjacent sides are $\vec{A} = 3\hat{i} - 2\hat{j} + 5\hat{k}$ and $\vec{B} = \hat{i} + 6\hat{j} - \hat{k}$.

16.2.2

Find the area of a parallelogram if $\vec{A} = \hat{i} - 4\hat{j} - \hat{k}$ and $\vec{B} = -2\hat{i} - \hat{j} + \hat{k}$ are the adjacent sides.

16.3 Parallelogram (Diagonals)

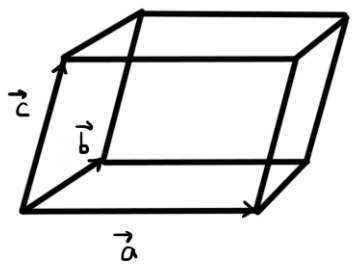
16.3.1

Find the area of a parallelogram whose diagonals are respectively $\vec{A} = 3\hat{i} + \hat{j} - 2\hat{k}$ and $\vec{B} = \hat{i} - 3\hat{j} + 4\hat{k}$.

17 Finding Volume of a Parallelepiped

17.0.1

Find the volume of the parallelepiped whose 3 sides are $\vec{a} = 3\hat{i} + \hat{j}, \vec{b} = \hat{j} + 3\hat{k}$ and $\vec{c} = -\hat{i} + 3\hat{j} + \hat{k}$.



18 Checking Coplanarity of three Vectors

18.0.1

If $\vec{A} = 2\hat{i} + \hat{j} - \hat{k}, \vec{B} = 3\hat{i} - 2\hat{j} + 4\hat{k}$ and $\vec{C} = \hat{i} - 3\hat{j} + 5\hat{k}$, show that the vectors are co-planar.

19 Relative Velocity

19.0.1

A man is moving with velocity $4ms^{-1}$ while rain falls with velocity $3ms^{-1}$ perpendicularly. At what angle does the man need to hold the umbrella with the horizontal to be safe from the rain?

19.0.2

Rain is falling with velocity $10km/h$ perpendicularly and wind blows with velocity $60km/h$ from the east to the west. Find the velocity of a car moving from east to west such that (a) front and rear windshield of the car get wet, (b) only the back windshield gets wet, (c) only the front windshield gets wet.

19.0.3

Two particles cross a point with velocities $12ms^{-1}$ and $20ms^{-1}$ at angle $120°$ with each other. What is the distance between them after $4s$?

19.0.4

A driver of a car moving with velocity $40km/h$ towards the east sees a truck moving with velocity $40\sqrt{3}km/h$ towards the north. (i) In which direction is the truck actually moving? (ii) What is the velocity of the truck?

19.0.5

A man is walking on a level road at a speed of $3.0km/h$. Rain drops fall vertically with a speed of $4.0km/h$. Find the velocity of the raindrops with respect to the man.

19.0.6

A man is moving with velocity $4m/s$ while the rain falls with velocity $3m/s$ perpendicularly. At what angle with the horizontal does the man need to hold the umbrella to be safe from the rain?

19.0.7

A man is moving with velocity $5km/h$ towards the west. Rain is falling with velocity $12km/h$ directly on his head. What is the true velocity of rain?

19.0.8

A man standing on a road has to hold his umbrella at $30°$ with the vertical to keep the rain away. He throws the umbrella and starts running at $10km/h$. He find that raindrops are hitting his head vertically. Find the speed of raindrops with respect to (a) the road, (b) the moving man.

19.0.9

A man running on a horizontal road at $8km/h$ finds the rain falling vertically. He increases his speed to $12km/h$ and finds that the drops make angle $30°$ with the vertical. Find the speed and direction of the rain with respect to the road.

19.0.10

A man moves with velocity $2ms^{-1}$ and later at a velocity $4ms^{-1}$ in the same direction. In the first case rain falls perpendicularly and in the second case at an angle $45°$ on him. Find the velocity and direction of the rain.

20 Numerical Problems Related to River and Boat

20.0.1

A river has got a current of $5kmh^{-1}$. If a boat is rowed at a velocity of $4kmh^{-1}$ making an angle $60°$ with the current, then actually what velocity and direction will the boat move?

20.0.2

A swimmer can swim in the river at a velocity of $3kmh^{-1}$ if there is no current. He started swimming with the intention of going straight from one side to another side of the river with current and finally crossed the river diagonally at the velocity of $5kmh^{-1}$. Calculate the velocity of the current of the river.

20.0.3

The velocity of a river current is $5kmh^{-1}$. A boat is steered with velocity $10kmh^{-1}$. At what direction does the boat need to be steered to cross the river directly?

20.0.4

A river has got a current of velocity $1kmh^{-1}$. A boatman is plying his boat at velocity of $3kmh^{-}1$ making an angle $30°$ with the current. If the width of the river is $0.5km$, calculate the time required to reach from one bank to the another.

20.0.5

A swimmer is trying to cross a $240m$ river. The velocity of the stream is $1800m$ per hour and the velocity of the swimmer is $3600m$ per hour. If he wants to cross the river perpendicularly, at what direction must he swim and how much time will it take for him?

20.0.6

The velocity of a boat is $18kmh^{-1}$ downstream and $6kmh^{-1}$ upstream. In which direction will the boat travel to cross the river directly and what is the resultant velocity of the boat?

20.0.7

The velocity of river current is $6kmh^{-1}$ and that of a boat is $12kmh^{-1}$. The width of the river is $10km$. (a) In which direction will the boat need to be steered to cross the river with minimum distance? What is the minimum distance? What is the minimum time? (b) In which direction will the boat need to be steered to cross the river with minimum time? What will be the distance travelled and time required?

20.0.8

A man can cross a river of width $100m$ directly in 4 minutes if there is no current and he takes 5 minutes if there is current. Find the velocity of the river current.

20.0.9

A thief standing on the bank of a river flowing with velocity $4ms^{-1}$ starts running along the bank of the river with constant velocity seeing a police boat on the directly opposite side of the river. The velocity of the police boat is $5ms^{-1}$ and is making angle $60°$ with the river current. If the police boat catches the thief, what was the velocity of the thief?

20.0.10

The velocities of the boat and the river current are $4ms^{-1}$ and $3ms^{-1}$ respectively. If the width of the river is $400m$, find the minimum time to cross the river. Also, find the resultant and direction of the boat.

20.0.11

Quvi bhai and Rafi bhai prepare for a boat race. Velocity of each boat is $10kmh^{-1}$. They start to cross a river of width $0.5km$. Cleverly Quvi bhai steers the boat at angle $120°$ with the current and Rafi bhai steers straight. [Velocity of river current is $5kmh^{-1}$].
(a) How far will Rafi bhai travel along the bank of the river from the opposite point of the starting position?
(b) "Despite being clever Quvi bhai will not win the race" - explain the statement.

21 Vector Calculus

21.0.1

If $\vec{A} = t^2\hat{i} - t\hat{j} + (2t+1)\hat{k}$ and $\vec{Q} = 5t\hat{i} + t\hat{j} - t^3\hat{k}$, find $\frac{d}{dt}(\vec{A}.\vec{Q})$ and $\frac{d}{dt}(\vec{A} \times \vec{Q})$.

22 Gradient of a Scalar Field

22.0.1

Find the gradient of the scalar field $\phi = x + xy + xyz$.

22.0.2

If $\phi(x, y, z) = 3xy^2z^3 - 4xy$, find $\vec{\nabla}\phi$ (grad ϕ). What is $\vec{\nabla}\phi$ at point $(2, -1, 1)$?

22.0.3

Find the gradient of ϕ if $\phi = \ln|\vec{r}|$, where $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ is the position vector of point (x, y, z).

22.0.4

If $\phi = 2xz^4 - x^2y$, find $\vec{\nabla}\phi$ and its magnitude at point $(2, -2, -1)$.

22.0.5

Find $\vec{\nabla}(\frac{1}{r})$, where $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ is the position vector of point (x, y, z).

23 Divergence of a Vector Field

23.0.1

If $\vec{A} = (3x^2z)\hat{i} + (xyz^2)\hat{j} - (x^3y^2z)\hat{k}$, find $\vec{\nabla}.\vec{A}$ at point $(1, -1, 1)$.

23.0.2

Find the divergence of the position vector $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$.

23.0.3

If $\vec{A} = 3xyz\hat{i} + 2xy^2\hat{j} - x^2yz\hat{k}$, find $\vec{\nabla}.\vec{A}$. What is the value at $(1, 1, -1)$?

23.0.4

If $\vec{r} = 3y^2z\hat{i} + 2x^3z\hat{j} - x^3y^2\hat{k}$, show that the vector field $\vec{r}$ is solenoidal.

23.0.5

For what value of b, is the vector $\vec{v} = (x + 3y)\hat{i} + (by - z)\hat{j} + (x - 2z)\hat{k}$ solenoidal?

24 Curl of a Vector Field

24.0.1

If $\vec{A} = (3x^2z)\hat{i} + (xyz^2)\hat{j} - (x^3y^2z)\hat{k}$, find $\vec{\nabla} \times \vec{A}$ at point $(1, -1, 1)$.

24.0.2

If $\vec{A} = 3xyz\hat{i} + 2xy^2\hat{j} - x^2yz\hat{k}$, find $\vec{\nabla} \times \vec{A}$ and $|\vec{\nabla} \times \vec{A}|$ at $(1, 1, -1)$?

24.0.3

Show that, the vector $\vec{A} = (x + 2y + 4z)\hat{i} + (2x - 3y - z)\hat{j} + (4x - y + 2z)\hat{k}$ is irrotational or conservative.